Mathematical Foundations

Worked solutions

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Exercise 1 — Set Equality

Let $A = \{2k \mid k \in \mathbb{Z}\}$ and $B = \{n \in \mathbb{Z} \mid n \text{ is even}\}$. We want to prove that $A = B$. It is sufficient to prove both inclusions, $A \subseteq B$ and $B \subseteq A$.

Proof that $A \subseteq B$

Let $x \in A$ be arbitrary. By the definition of A, there exists some $k \in \mathbb{Z}$ such that $x = 2k$. An integer is even precisely when it has this form, so x is even. Since $x \in \mathbb{Z}$, it follows that $x \in B$. As x was arbitrary, $A \subseteq B$.

Proof that $B \subseteq A$

Let $x \in B$ be arbitrary. By the definition of B, x is an even integer, so there exists some $k \in \mathbb{Z}$ such that $x = 2k$. Every integer of this form belongs to A; hence $x \in A$. As x was arbitrary, $B \subseteq A$. The two inclusions together prove that

$$\boxed{A = B}.$$

□

Exercise 2 — Power Sets

Let $A = \{a, b, c\}$. The power set $\mathcal{P}(A)$ contains every subset of A, so

$$\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

There are eight listed subsets. Since $|A| = 3$, we also have $2^{|A|} = 2^3 = 8$. Therefore,

$$\boxed{|\mathcal{P}(A)| = 2^{|A|}}.$$

The general reason for this formula is that, when constructing a subset, every element has exactly two possibilities: it is either included or excluded. Thus, if a set contains n elements, the number of possible subsets is

$$\underbrace{2 \cdot 2 \cdots 2}_{n \text{ times}} = 2^n.$$

Exercise 3 — Cartesian Products

Let $A = \{0, 1\}$ and $B = \{a, b, c\}$.
By definition,

$$A \times B = \{(x, y) \mid x \in A,\ y \in B\}.$$

Therefore,

$$A \times B = \{(0, a), (0, b), (0, c), (1, a), (1, b), (1, c)\}.$$

Similarly,

$$B \times A = \{(x, y) \mid x \in B,\ y \in A\},$$

so

$$B \times A = \{(a, 0), (a, 1), (b, 0), (b, 1), (c, 0), (c, 1)\}.$$

These sets are not equal: for example, $(0, a) \in A \times B$, but $(0, a) \notin B \times A$. Therefore,

$$\boxed{A \times B \neq B \times A}.$$

Although $|A \times B| = |B \times A|$, equal cardinality does not imply equality of sets.

Exercise 4 — Logical Implication

We analyze each implication for arbitrary $n \in \mathbb{Z}$.

Implication 1

Suppose $4 \mid n$. By the definition of divisibility, there exists some $k \in \mathbb{Z}$ such that

$$n = 4k = 2(2k).$$

Since $2k \in \mathbb{Z}$, it follows that $2 \mid n$. Therefore,

$$\boxed{4 \mid n \implies 2 \mid n}$$

is true for every $n \in \mathbb{Z}$.

Implication 2

The implication is false. Take $n = 2$: we have $2 \mid 2$ because $2 = 2 \cdot 1$, but $4 \nmid 2$. Thus $n = 2$ is a counterexample, and

$$\boxed{2 \mid n \implies 4 \mid n}$$

is false in general.

Implication 3

Suppose $6 \mid n$. Then there exists some $k \in \mathbb{Z}$ such that

$$n = 6k = 3(2k).$$

Because $2k \in \mathbb{Z}$, this proves that $3 \mid n$. Therefore,

$$\boxed{6 \mid n \implies 3 \mid n}$$

is true for every $n \in \mathbb{Z}$.

Exercise 5 — Contrapositive

The original implication is

$$3 \mid n \implies n^2 \equiv 0 \pmod 3.$$

The contrapositive of $P \implies Q$ is $\neg Q \implies \neg P$. Here, P is the statement $3 \mid n$, and Q is the statement $n^2 \equiv 0 \pmod 3$. Therefore, the contrapositive is

$$\boxed{n^2 \not\equiv 0 \pmod 3 \implies 3 \nmid n}.$$

Exercise 6 — Quantifier Negation

The original statement is

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, \quad y > x.$$

Negating the entire statement gives

$$\neg \left(\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, \quad y > x \right).$$

Using

$$\neg(\forall x, P(x)) \iff \exists x, \neg P(x),$$

we obtain

$$\exists x \in \mathbb{Z}, \neg \left(\exists y \in \mathbb{Z}, \quad y > x \right).$$

Next, using

$$\neg(\exists y, P(y)) \iff \forall y, \neg P(y),$$

we obtain

$$\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, \quad \neg(y > x).$$

Since

$$\neg(y > x) \iff y \leq x,$$

the negation is

$$\boxed{\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, \quad y \leq x}.$$

Exercise 7 — Quantifier Order

Consider first

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad x + y = 0.$$

Let $x \in \mathbb{R}$ be arbitrary and choose $y = -x$. Since $-x \in \mathbb{R}$,

$$x + y = x + (-x) = 0.$$

Therefore,

$$\boxed{\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \quad x + y = 0}$$

is true.
Now consider

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad x + y = 0.$$

This statement claims that a single real number y satisfies $x + y = 0$ for every $x \in \mathbb{R}$. Suppose such a y exists. Taking $x = 0$ gives $y = 0$, while taking $x = 1$ would then require $1 + 0 = 0$, which is false.

Therefore,

$$\boxed{\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, \quad x + y = 0}$$

is false.

The difference lies in the dependency of the quantified variables. In $\forall x \exists y$, the value of y may depend on x; here, we choose $y = -x$. In $\exists y \forall x$, a single fixed value of y must work for every possible x.

Thus, in general,

$$\forall x \exists y \not\equiv \exists y \forall x.$$

Exercise 8 — Relations

Define a relation R on $\mathbb{Z}$ by

$$aRb \iff a - b \text{ is even.}$$

We prove that R is reflexive, symmetric, and transitive.

Reflexivity

Let $a \in \mathbb{Z}$ be arbitrary. Since $a - a = 0 = 2 \cdot 0$, the difference $a - a$ is even, so aRa. Thus R is reflexive.

Symmetry

Suppose aRb. Then $a - b$ is even, so there exists some $k \in \mathbb{Z}$ such that

$$a - b = 2k.$$

Then

$$b - a = -(a - b) = -2k = 2(-k).$$

Since $-k \in \mathbb{Z}$, the difference $b - a$ is even and hence bRa. Thus R is symmetric.

Transitivity

Suppose aRb and bRc. Then there exist $k, l \in \mathbb{Z}$ such that

$$a - b = 2k$$

and

$$b - c = 2l.$$

Adding both equations,

$$(a - b) + (b - c) = 2k + 2l.$$

Hence,

$$a - c = 2(k + l).$$

Since $k + l \in \mathbb{Z}$, the difference $a - c$ is even and hence aRc. Thus R is transitive.
Since R is reflexive, symmetric, and transitive,

$$\boxed{R \text{ is an equivalence relation.}}$$

Equivalence Classes

The equivalence class of an integer a is

$$[a] = \{x \in \mathbb{Z} \mid xRa\}.$$

By the definition of R,

$$[a] = \{x \in \mathbb{Z} \mid x - a \text{ is even}\}.$$

There are exactly two distinct equivalence classes.
The equivalence class of 0 is

$$[0] = \{x \in \mathbb{Z} \mid x \text{ is even}\}.$$

Thus,

$$[0] = \{\ldots, -4, -2, 0, 2, 4, \ldots\}.$$

Equivalently,

$$[0] = 2\mathbb{Z}.$$

The equivalence class of 1 is

$$[1] = \{x \in \mathbb{Z} \mid x - 1 \text{ is even}\}.$$

This is precisely the set of odd integers:

$$[1] = \{\ldots, -3, -1, 1, 3, 5, \ldots\}.$$

Equivalently,

$$[1] = 2\mathbb{Z} + 1.$$

Since every integer is either even or odd, these are the only two distinct equivalence classes.
Therefore,

$$\boxed{\mathbb{Z}/R = \{[0], [1]\}}.$$

Exercise 9 — Injectivity and Surjectivity

Consider the function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = 3x + 1$.

Injectivity

Suppose $f(x_1) = f(x_2)$. Then

$$3x_1 + 1 = 3x_2 + 1.$$

Subtracting 1 and dividing by 3 gives $x_1 = x_2$. Therefore,

$$\boxed{f \text{ is injective.}}$$

Surjectivity

For f to be surjective, every $y \in \mathbb{Z}$ would need a preimage $x \in \mathbb{Z}$ satisfying $3x + 1 = y$. Solving for x gives

$$x = \frac{y - 1}{3}.$$

This expression is not necessarily an integer. For example, when $y = 0$ the only possible value is $x = -\frac{1}{3} \notin \mathbb{Z}$, so 0 has no preimage under f.

Therefore,

$$\boxed{f \text{ is not surjective.}}$$

Its image is

$$\mathrm{Im}(f) = \{3k + 1 \mid k \in \mathbb{Z}\}.$$

Equivalently,

$$\mathrm{Im}(f) = \{n \in \mathbb{Z} \mid n \equiv 1 \pmod 3\}.$$

Hence, f is injective but not surjective, and therefore it is not bijective.

Exercise 10 — Dependence on the Codomain

Consider the functions

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2, \qquad g : \mathbb{R} \to [0, \infty), \quad g(x) = x^2.$$

Function f

Injectivity

The function f is not injective because $f(1) = f(-1) = 1$ while $1 \neq -1$. Therefore,

$$\boxed{f \text{ is not injective.}}$$

Surjectivity

For every $x \in \mathbb{R}$, we have $x^2 \geq 0$. Therefore, no negative real number belongs to the image of f; for example, there is no real solution to $x^2 = -1$.

Conversely, let $y \in [0, \infty)$ be arbitrary and choose $x = \sqrt{y}$. Then $x \in \mathbb{R}$ and $f(x) = x^2 = y$, so every nonnegative real number belongs to the image. Hence both inclusions hold, and

$$\mathrm{Im}(f) = [0, \infty) \neq \mathbb{R}.$$

Therefore,

$$\boxed{f \text{ is not surjective.}}$$

Function g

Injectivity

As before, $g(1) = g(-1) = 1$ with $1 \neq -1$. Therefore,

$$\boxed{g \text{ is not injective.}}$$

Surjectivity

Let $y \in [0, \infty)$ be arbitrary. Since $y \geq 0$, choose the real number $x = \sqrt{y}$. Then

$$g(x) = x^2 = (\sqrt{y})^2 = y.$$

Thus every element of the codomain has a preimage, and therefore

$$\boxed{g \text{ is surjective.}}$$

The two functions have the same formula $x \mapsto x^2$ but different codomains. For f, the image $[0, \infty)$ is a proper subset of the codomain $\mathbb{R}$, so f is not surjective. For g, the same image is exactly the codomain, so g is surjective. This demonstrates that surjectivity is a property of the entire function, including its domain and codomain, rather than only of its formula.

Exercise 11 — Restricted Domains

Define $f : [0, \infty) \to [0, \infty)$ by $f(x) = x^2$. We prove that f is both injective and surjective.

Injectivity

Suppose $f(x_1) = f(x_2)$. Subtracting and factoring gives

$$(x_1 - x_2)(x_1 + x_2) = 0.$$

Thus $x_1 = x_2$ or $x_1 = -x_2$. Because both values are nonnegative, the second equality can occur only when $x_1 = x_2 = 0$. In every case $x_1 = x_2$, and therefore

$$\boxed{f \text{ is injective.}}$$

Surjectivity

Let $y \in [0, \infty)$ be arbitrary and choose $x = \sqrt{y}$. Since $y \geq 0$, this choice belongs to the domain, and

$$f(x) = x^2 = (\sqrt{y})^2 = y.$$

Therefore,

$$\boxed{f \text{ is surjective.}}$$

Since f is both injective and surjective,

$$\boxed{f \text{ is bijective.}}$$

To determine the inverse, solve $y = x^2$ for x. Because the domain is restricted to nonnegative values, $x = \sqrt{y}$. Hence,

$$\boxed{f^{-1}(x) = \sqrt{x}}.$$

Indeed, $f^{-1}(f(x)) = \sqrt{x^2} = x$ and $f(f^{-1}(x)) = (\sqrt{x})^2 = x$ for every $x \in [0, \infty)$.

Exercise 12 — Composition

Let

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x + 1,$$

and

$$g : \mathbb{R} \to \mathbb{R}, \qquad g(x) = x^2.$$

By definition, $(g \circ f)(x) = g(f(x))$. Since $f(x) = 2x + 1$, we obtain

$$(g \circ f)(x) = g(2x + 1) = (2x + 1)^2.$$

Therefore,

$$\boxed{(g \circ f)(x) = 4x^2 + 4x + 1}.$$

Similarly, $(f \circ g)(x) = f(g(x))$. Since $g(x) = x^2$, we obtain

$$(f \circ g)(x) = f(x^2) = 2x^2 + 1.$$

Therefore,

$$\boxed{(f \circ g)(x) = 2x^2 + 1}.$$

These functions are not equal: evaluating at $x = 1$ gives $(g \circ f)(1) = 9$ while $(f \circ g)(1) = 3$. Therefore,

$$\boxed{g \circ f \neq f \circ g}.$$

This illustrates that function composition is generally not commutative.

Exercise 13 — Direct Proof

We prove that the sum of two odd integers is even.

Proof

Let $a, b \in \mathbb{Z}$ be arbitrary odd integers.

By the definition of an odd integer, there exist $k, l \in \mathbb{Z}$ such that $a = 2k + 1$ and $b = 2l + 1$. Therefore,

$$a + b = 2(k + l + 1).$$

Since $k, l \in \mathbb{Z}$ and the integers are closed under addition, $k + l + 1 \in \mathbb{Z}$. Thus $a + b$ has the form $2m$ for an integer m.

By the definition of evenness,

$$\boxed{a + b \text{ is even.}}$$

□

Exercise 14 — Contraposition

We want to prove that, for every $n \in \mathbb{Z}$,

$$n^2 \text{ odd} \implies n \text{ odd}.$$

We prove the contrapositive.
The contrapositive is

$$n \text{ not odd} \implies n^2 \text{ not odd}.$$

Since every integer is either even or odd, this is equivalent to

$$n \text{ even} \implies n^2 \text{ even}.$$

Proof

Let $n \in \mathbb{Z}$ be even. By definition, there exists some $k \in \mathbb{Z}$ such that $n = 2k$. Then

$$n^2 = 2(2k^2).$$

Since $2k^2 \in \mathbb{Z}$, the square n^2 is even.
Thus,

$$n \text{ even} \implies n^2 \text{ even}.$$

We have proved the contrapositive. Therefore,

$$\boxed{n^2 \text{ odd} \implies n \text{ odd}}.$$

□

Exercise 15 — Correctness Implies Injectivity

Let $E : M \to C$ and $D : C \to M$ be functions satisfying $D(E(m)) = m$ for every $m \in M$.
Equivalently,

$$D \circ E = \mathrm{id}_M .$$

We first prove that E is injective.

Injectivity of E

Let $m_1, m_2 \in M$ and suppose $E(m_1) = E(m_2)$. Applying D to both sides gives

$$D(E(m_1)) = D(E(m_2)).$$

By the assumed correctness property, the left-hand side is m_1 and the right-hand side is m_2. Therefore $m_1 = m_2$, and hence

$$E(m_1) = E(m_2) \implies m_1 = m_2.$$

By definition,

$$\boxed{E \text{ is injective.}}$$

□

Is D necessarily injective?

No. Consider $M = \{0\}$ and $C = \{a, b\}$. Define $E(0) = a$ and define $D : C \to M$ by $D(a) = D(b) = 0$. Then

$$D(E(0)) = D(a) = 0.$$

Thus $D(E(m)) = m$ for every $m \in M$. However, $a \neq b$ while $D(a) = D(b) = 0$, so D is not injective.

Hence,

$$\boxed{D \text{ is not necessarily injective.}}$$

The condition $D \circ E = \mathrm{id}_M$ does imply two general properties: E is injective and D is surjective.

To see why D is surjective, let $m \in M$ be arbitrary. The element $E(m) \in C$ satisfies $D(E(m)) = m$, so every element of M has a preimage under D.

In general, however, the condition does not imply that E is surjective or that D is injective.